

Seohyeon Cha

Ph.D. Student in Electrical and Computer Engineering

Austin, Texas | seohyeon.cha@utexas.edu | +1 512-992-4955

[Website](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

Research Profile

My research focuses on **resource-efficient machine learning** for deploying and adapting AI models in distributed and hardware-constrained environments. I develop algorithms that reduce the memory, computation, and communication costs of both training and inference. This work addresses a central barrier to practical AI deployment: the growing resource demands of modern AI models limit where they can operate, how frequently they can be updated, and how effectively distributed devices can collaborate. My recent research advances this goal by developing methods for model compression, distributed and continual adaptation, and resource-aware training and inference across cloud, edge, and embedded systems.

Education

The University of Texas at Austin

Ph.D. in Electrical and Computer Engineering; GPA: 4.00/4.00

Austin, TX

2024–Present

- Advisor: Prof. Haris Vikalo

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Electrical Engineering; GPA: 4.17/4.30

Republic of Korea

2022–2024

- Advisor: Prof. Joonhyuk Kang

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Electrical Engineering; GPA: 4.03/4.30; Summa Cum Laude

Republic of Korea

2017–2022

Research Experience

Graduate Research Assistant

Wireless Networking and Communications Group, The University of Texas at Austin

Austin, TX

2024–Present

- Developed post-training quantization methods that improve low-bit LLM inference by correcting error propagation across quantized layers without retraining the model.
- Designed memory- and communication-efficient methods for continual and federated learning, including replay-free gradient projection, compressed task subspaces, and aggregation of low-rank adapters.
- Studied model placement and workload routing for hierarchical device–edge–cloud inference under memory, computation, communication, and latency constraints.

Graduate Researcher

Advanced Radio Technology Laboratory, KAIST

Daejeon, Republic of Korea

2022–2024

- Developed federated-learning frameworks that enable clients with heterogeneous model architectures and computational capabilities to exchange knowledge through nested model scaling and generative models.
- Studied uncertainty quantification for graph neural networks and optimization of intelligent surface-aided millimeter-wave communication systems.

Industry Experience

Kilby Labs, Texas Instruments
Research Intern, Embedded AI Team

Dallas, TX
Summer 2026

- Investigated memory-efficient on-device training and continual adaptation of quantized neural networks for microcontrollers.
- Developed fixed-point training and selective-update strategies to reduce training memory and computation under strict device resource constraints.

SK hynix
Undergraduate Summer Intern

Summer 2019

Peer-Reviewed Publications

*Boldface identifies Seohyeon Cha. *Equal contribution.*

- [1] **S. Cha**, H. Chen, D. Kim, H. Zhang, K. S. Chan, G. de Veciana, and H. Vikalo, “CoreQ: Learning-Free Mismatch Correction and Successive Rounding for Quantization,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Spotlight (top 5.1% of accepted papers)**.
- [2] H. Zhang, D. Kim, **S. Cha**, and H. Vikalo, “FedRot-LoRA: Mitigating Rotational Misalignment in Federated LoRA,” in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.
- [3] D. Kim, **S. Cha**, H. Chen, C. Wang, and H. Vikalo, “Quantized Gradient Projection for Memory-Efficient Continual Learning,” in *Proceedings of the 14th International Conference on Learning Representations (ICLR)*, 2026.
- [4] **S. Cha***, H. Chen*, and H. Vikalo, “FedProTIP: Task-Agnostic Federated Continual Learning via Replay-Free Gradient Projection,” *Transactions on Machine Learning Research*, 2026.
- [5] **S. Cha**, K. S. Chan, G. de Veciana, and H. Vikalo, “Joint Model Onloading and Offloading for Hierarchical Multi-Task Inference,” in *Proceedings of the IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, 2026.
- [6] H. Kang*, **S. Cha***, and J. Kang, “GeFL: Model-Agnostic Federated Learning With Generative Models,” *IEEE Transactions on Mobile Computing*, vol. 25, no. 4, pp. 4493–4505, 2026.
- [7] H. Kang, **S. Cha**, J. Shin, J. Lee, and J. Kang, “NeFL: Nested Model Scaling for Federated Learning With System Heterogeneous Clients,” *IEEE Transactions on Mobile Computing*, vol. 24, no. 8, pp. 6734–6746, 2025.
- [8] **S. Cha**, H. Kang, and J. Kang, “On the Temperature of Bayesian Graph Neural Networks for Conformal Prediction,” in *NeurIPS 2023 Workshop on New Frontiers in Graph Learning*, 2023.
- [9] **S. Cha**, S. Kim, J. Seo, and J. Kang, “Intelligent Surface-Aided Transmit-Array Antenna in mmWave Communication System with Historical Channel Observation,” in *Proceedings of the IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 2022.

Preprints

- [10] H. Zhang, **S. Cha**, H. B. Beytur, K. S. Chan, G. de Veciana, and H. Vikalo, “Online Learning for Multi-Layer Hierarchical Inference under Partial and Policy-Dependent Feedback,” arXiv:2603.04247, 2026.

Honors and Awards

Best Project Award <i>Machine Learning on Real-World Networks, The University of Texas at Austin</i> <i>Project: Distributed Continual Learning Using Gradient Projection</i>	2024
Korean Government Scholarship <i>KAIST Graduate Program</i>	2022–2024
National Science and Engineering Scholarship <i>Academic Excellence</i>	2019–2021
Korean Government Scholarship <i>KAIST Undergraduate Program</i>	2017–2018

Teaching and Mentoring

Undergraduate Individual Study Assistant, KAIST Guided students in machine-learning theory and implementation, including federated learning.	2023
Teaching Assistant, KAIST EE205 Data Structures and Algorithms for Electrical Engineering EE966 M.S. Seminar / Colloquium.	2022–2023
Counseling Assistant, KAIST Provided academic counseling and peer support to undergraduate and graduate students.	Sep 2022–Feb 2023
Freshman Tutor, KAIST MAS101 Calculus I and MAS102 Calculus II.	2018–2019

Technical Skills

Programming	Python, C++, MATLAB, $\text{\LaTeX}$
ML frameworks	PyTorch, TensorFlow, Hugging Face
Model optimization	Post-training quantization, quantization-aware training, fixed-point/integer training, parameter-efficient fine-tuning
Tools	Linux, Git, Pandas
Languages	Korean (native), English (proficient)